STAT 330-001

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SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE



EXCEPTION



NS

STAT 330 - Exam 1

073

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO and ridge regression are called "shrinkage" methods because they are ~~reducing the computational~~ underemphasizing the effects of many predictors's on the response (or omitting them altogether) with the goal of creating a more parsimonious model. To do this, shrinkage methods push coefficients for predictors closer to 0, understating their effect, and, in the case of LASSO, can even set coefficients exactly to 0.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^TX)^{-1}(X^TX)Y$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is a statistic and point estimate for the true population parameter β. $\hat{\beta}$ is calculated from our sample to estimate β.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

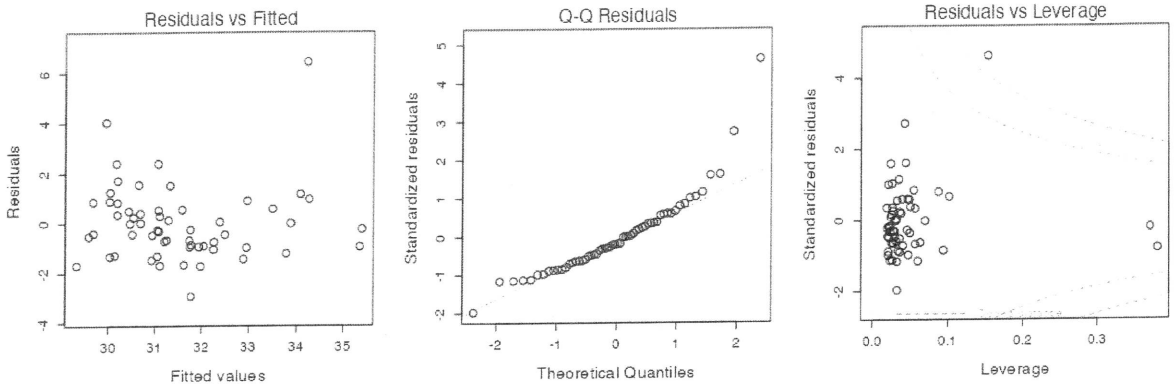
We use properties of linear algebra such as invertible matrices and "normal" matrices to algebraically solve for a vector that contains regression coefficients, which estimate the effect of changing levels of predictors

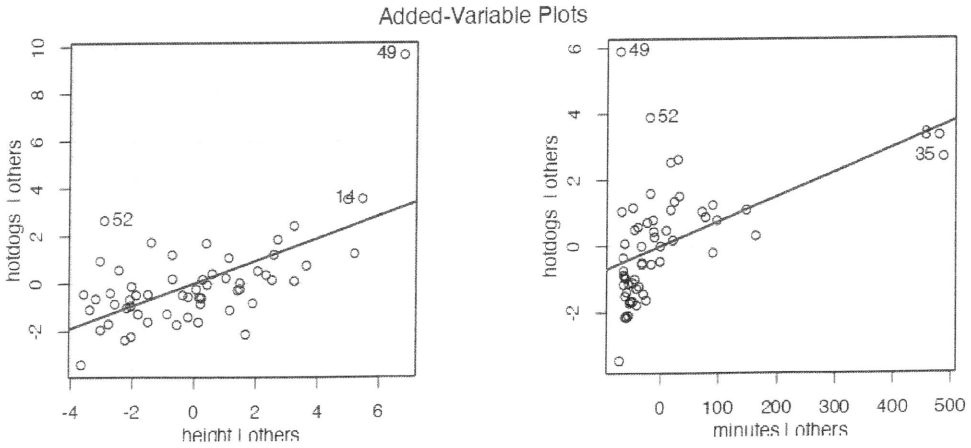
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1. and interpret its value in the context of the problem

The estimated value of β_1 is 0.472892. It means that, holding all other predictors constant, an increase of height by 1 inch for a contestant is associated with eating an estimated 0.473 hotdogs, on average.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$ — NOT selected. Sorry my eraser is bad.

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Based on the AV plots, it would appear that the relationship between height and hotdogs eaten is linear, while the relationship between minutes trained per week and hotdogs eaten does not appear to be linear.

(B) There is independence between each observation

It's hard to tell for sure, but based on the way the data was collected, I would say that independence of observations is a fair assumption. It is unfortunate that so many are from the same 4 countries, though.

(C) The errors are normally distributed

Based on the QQ plot we see that the tails of the distribution of our residuals does not match the normal model well, but I would say that the overall fit is pretty close and it is ok to proceed under the normal errors assumption.

(D) The errors have constant variance (homoscedasticity)

Based on the residual vs fitted plot, we do see slightly more spread out residuals on the left side of the graph, but overall I would conclude that the assumption of constant variance can be judged as appropriate here.

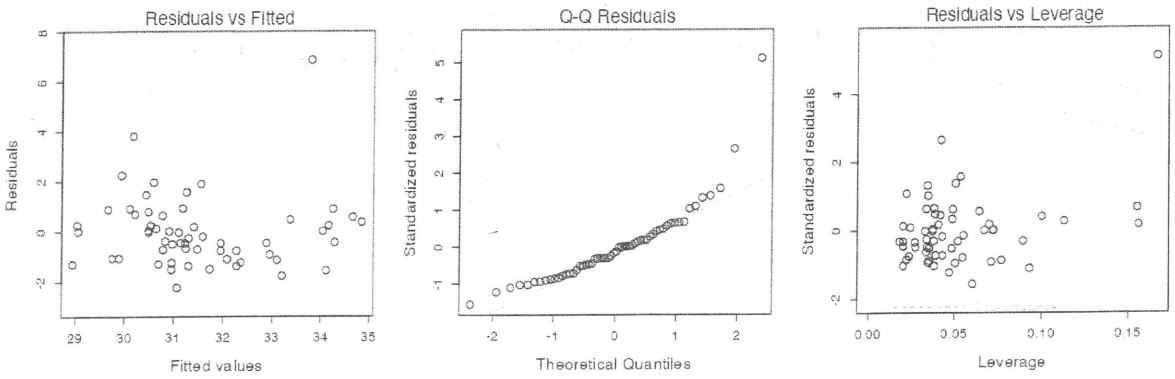
(E) There are no unduly influential points

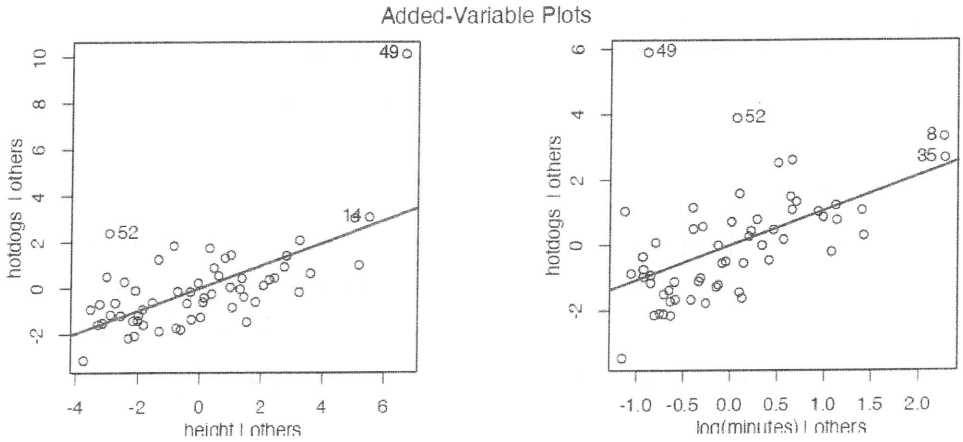
Based on the leverage plot, it does not appear that our assumption of no unduly influential points is met here. I'd be concerned about the two points on the far right and the one that's very high in the middle.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would respond that making sure that all of our assumptions are met for the model we're using is what is most critical and will allow our statistical work to be ethical and function properly, not what gets the best p-values. I'd explain to them that the linearity assumption for minutes and hotdogs is greatly improved here and that the model seems more appropriate.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 seems to follow assumptions much better. I'm quite happy with the model. one downside is that the coefficient for log(minutes) becomes less interpretable and could become confusing for those looking at our model.

Question 16: State the estimated value of β_2 and provide an interpretation in context

The estimated value of β_2 is 1.06702. It means that, holding all other predictors constant, an increase of 1 in log(minutes framed) is associated with eating 1.06702 additional hotdogs, on average.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.05589 + 68(0.46834) + 1.06702(4.369) =$

31.81 hotdogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes. what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$0.204 \cdot 2.004 = 0.4086$

$1.485 \cdot 2.004 = 2.97594$

$31.81 + 0.4086, \quad 31.81 + 2.97594$

$(31.81 - 0.4086, \quad 31.81 + 2.97594)$

Part III - Application

Questions 20-24 concern the following dataset:

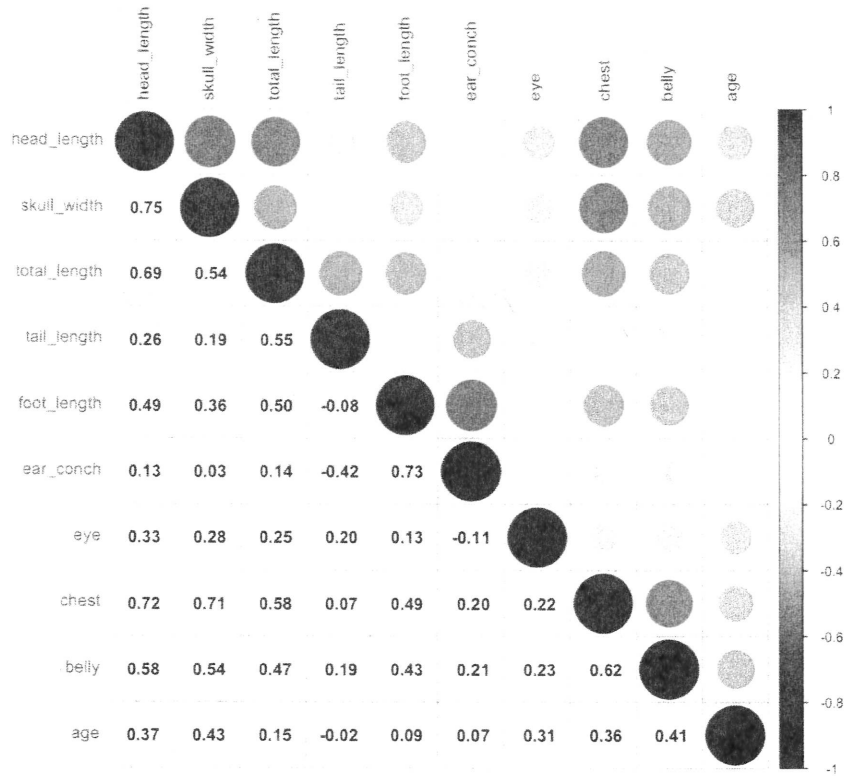
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that, among the population of all male local Australian possums, an increase of 1cm of belly girth is associated with ~~somewhere~~ anywhere from a -0.023 to 0.443 year difference in age, on average.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The F test tests whether all the predictors have a significant effect, while t test tests if their coefficients are not equal to 0

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

This algorithm creates all possible models and decides one the best one based on which has the lowest AIC

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

Scratch Paper

Scratch Paper

Scratch Paper

Scratch Paper